

ANDRÉ GAMA DE MATOS

ELECTRONICS TEST & R&D ENGINEER | EMBEDDED SYSTEMS | TEST AUTOMATION |
COMPUTER VISION & INDUSTRIAL AI

Manaus, Amazonas, Brazil | andrematos37@hotmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Electronics Test & R&D Engineer with 26 years of experience in the electronics industry across Product Engineering, Test Engineering, NPI, Reliability, Failure Analysis and technology development. Experienced in the design and deployment of industrial Computer Vision, test automation, embedded systems and IIoT solutions. Currently working in two parallel R&D fronts: Project Engineering and Test Engineering at Instituto Conecthus, supporting new TV, Memory and SSD products, and as a Resident Researcher in Microelectronics at CBPF, with SystemVerilog, RTL, verification, communication protocols, Cadence tools and PDK-based flows. Fluent English for technical communication and global collaboration.

CORE SKILLS

Test Engineering / PCBA: ICT/FCT, functional test, test fixtures and programs, troubleshooting, CPK, FMEA, reliability and NPI.

Computer Vision / AI: OpenCV, TensorFlow/Keras, CNNs, MobileNetV2, ROI processing, datasets, validation and deployment.

Product / Manufacturing: TV, Memory and SSD validation, hardware/firmware, NPI, BIOS, drivers, SAP/BOM, ECO/ECR and manufacturing.

Embedded / Automation: Python, C/C++, ESP32, Arduino, industrial sensors, serial communication, IoT/IIoT and web interfaces.

Digital Design / Microelectronics: SystemVerilog, RTL, testbenches, SPI, ASIC flow, Cadence Xcelium, synthesis, timing and PDK.

R&D / Documentation: Proofs of concept, experimental validation, root-cause analysis, reports, manuals, releases and technology transfer.

SELECTED PROJECTS

SAMSP - Smart Ambient Monitoring for Solder Paste

Industrial IoT solution for temperature/humidity monitoring, alarms, door status, QR Code traceability, FIFO inventory and web supervision. Developed in Python/C++ with ESP32; INPI-registered software (BR512026007033-3). [GitHub](#) | [DOI](#)

SVC USB - Multi-product automated USB connector inspection

Computer Vision system for USB-A and USB-C using Deep Learning, QR Code triggering, model recipes, traceability and yield indicators. INPI-registered software. [GitHub](#) | [DOI](#)

SVC Springs - Dual spring inspection platform

Industrial system using Python, OpenCV, TensorFlow/Keras, Streamlit, USB camera, sensor and Arduino; validated with more than 1,000 units. M.Sc. project and INPI-registered software. [GitHub](#) | [DOI](#)

SAPA IoT - ESP32-based industrial automation platform

IIoT system for autonomous control and energy management of industrial audio equipment, with scheduling, web administration, status logs and field deployment. [GitHub](#) | [DOI](#)

SPI Master Controller - Private Microelectronics Residency Project

Development and verification of an 8-bit full-duplex SPI Master in SystemVerilog, including RTL, testbench and Cadence Xcelium simulation. Source code remains private due to confidentiality.

PROFESSIONAL EXPERIENCE

Instituto Conecthus - Test Engineering Analyst - Project Engineering

Aug/2026 - Present

Manaus-AM | On-site

- Test Engineering projects for new TV, Memory and SSD products, supporting validation, troubleshooting, product introduction and test processes.
- Development and improvement of test and automation solutions using Python, C/C++, Arduino/ESP32, sensors, embedded systems, data acquisition and Computer Vision.

Brazilian Center for Research in Physics (CBPF) - Resident Researcher in Microelectronics

Jul/2026 - Present

Remote / Manaus-AM

- Research and development in digital circuit design and verification using SystemVerilog, RTL, testbenches, communication protocols, Cadence Xcelium and PDK-based IC flows.
- Developed and verified an 8-bit SPI Master controller and contributes to microelectronics activities applied to imaging systems.

Salcomp PLC - Senior Product Engineering Analyst

Mar/2023 - Jul/2026

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- Product Engineering, Test, Reliability and R&D for smart chargers and switched-mode power supplies for smartphones, tablets, notebooks and POS terminals.
- Developed test programs/fixtures, supported NPI and functional validation, performed circuit/failure analysis, reliability investigations, CPK evaluation and troubleshooting.
- Designed and deployed Industrial AI and IIoT solutions integrating cameras, sensors, Arduino/ESP32, software, traceability and production dashboards.

Positivo Tecnologia - Senior Product Analyst

Sep/2022 - Dec/2022

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- Product Engineering, Test and Technology Validation for notebooks, desktops, tablets, smartphones and POS terminals; technical coordination of validation and automation activities.
- Development and maintenance of scripts, BIOS, drivers, Windows/Linux images, operating systems, IMEI keys, server infrastructure and technical documentation.

Western Digital - Engineering Specialist - NPI & Technology Validation

Nov/2011 - Jun/2022

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- Led Product Engineering and NPI activities for HDDs, including software, firmware, test-system and industrial-infrastructure validation.
- Supported servers, databases and process-control systems and worked with global teams in the U.S. and Asia on product introduction, reliability and failure reduction.

Digitron - Product Engineering Analyst

Oct/2010 - Nov/2011

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- Product Engineering and NPI for HDDs, including test-system validation, failure analysis, firmware/software updates and manufacturing support.

Samsung Electronics - Process Engineering Technician (Jr / Mid / Sr)

Jul/2004 - Oct/2010

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- Product Engineering, Process Engineering and NPI for TVs, DVD systems, audio products, MP3 players and monitors; hardware/firmware validation, prototype analysis and test-equipment updates.
- BOM/SAP and ECO/ECR management and collaboration with global R&D in South Korea on product introduction, circuit analysis and technical problem solving.

Panasonic - Quality Technician / Product Validation

Feb/2001 - Jun/2004

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- Product validation and Quality Engineering for TVs, including functional testing, embedded hardware/software validation, sample analysis and technical documentation.

Elsys - Electronics Technician Intern

Jan/2000 - Oct/2000

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- Diagnosis, validation and repair of pay-TV decoders and electronic boards using oscilloscopes, spectrum/audio analyzers, multimeters, frequency counters and adjustable AC sources.

EDUCATION & TRAINING

Professional M.Sc. in Electrical Engineering - Embedded Systems, UEA | 2024 - 2026, in progress

B.Sc. in Control and Automation Engineering, UniFatecie | 2023 - 2026, in progress

B.Sc. in Software Engineering, UNIASSELVI | completed in 2025

Postgraduate Specialization in Automated Test Development for Intelligent Systems, UEA | 2024

Postgraduate Specialization in Software Engineering and Architecture, PUC Minas | 2023

Postgraduate Specialization in Electronics and Computer Engineering, Unyleya | 2023

Technology Degree in Industrial Automation, UNIP | 2014

Electronics Technician Diploma, IFAM | 2000

Microelectronics Training Program, CI Amazônia / iRede | 320 hours, completed Jul/2026, final grade 9.60

Additional Training: FMEA - Failure Mode and Effects Analysis | Metallography | Advanced SMT for Industry 4.0 | AATEC DAY 2025 - Artificial Intelligence.

LANGUAGES & INTELLECTUAL PROPERTY

Portuguese: native | English: fluent | Spanish: limited professional communication.

4 software registrations at INPI, including SVC Springs - BR512026003433-7 | SVC USB - BR512026005765-5 | SAPA IoT | SAMSP - BR512026007033-3.